



Intelligibility Prediction for Time-Modified Speech Signals Using Spectro-Temporal Modulation Features

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Abstract

Reference-based speech intelligibility prediction algorithms (RB-SIPAs) are limited to speech degradations that maintain the time alignment between the clean and degraded speech signals. We address this limitation by augmenting the existing RB-SIPAs framework with time alignment. To achieve robust time alignment at low signal-to-noise ratios, we propose using spectro-temporal modulations (STM) of the speech signals for dynamic time warping (DTW). Our experiments demonstrate that in DTW, the use of specific STM components as features for clean and time-modified *degraded* speech achieves the baseline time alignment obtained using clean speech and *noise-free* time-modified speech. Moreover, we propose two methods to incorporate time alignment into existing RB-SIPAs. Using these methods, we compare the output scores of the RB-SIPA with listening test scores and show better correlation results using the STM features as compared to MFCCs.

Index Terms: speech intelligibility prediction, dynamic time warping, spectro-temporal modulation

1. Introduction

Speech intelligibility prediction (SIP) involves predicting the degree to which normal hearing listeners identify a spoken message. SIP algorithms (SIPAs) are designed to replace time-consuming and costly listening tests for assessing speech intelligibility (SI) under various degradation conditions. Reference-based SIP algorithms (RB-SIPAs) typically entail measuring the similarity between a clean reference signal and the corresponding degraded signal, by comparing features crucial for understanding speech. Although effective for assessing SI, existing RB-SIPAs [1–4] assume time alignment of the input signals. This limits their utility, as the time dimension of the degraded speech signal may have been modified to suit application needs, such as pre-degradation speech enhancement (SE) [5, 6], speaking style change [7], etc. On the other hand, reference-free SIPAs (RF-SIPAs) require only the degraded speech signal to generate SIP [8]. However, RB-SIPAs and RF-SIPAs are complementary. For instance, RB-SIPAs have been used to label degraded signals for training RF-SIP deep neural networks (DNNs) [9]. RB-SIPAs have also been used as loss measures for training SE algorithms [10]. Since SE DNNs are capable of altering aspects of spoken speech such as making phoneme or speaker identity substitution [11], the lack of a reference signal makes it harder for RF-SIPAs to identify signals with fake constituents. Therefore, RB-SIPAs that could allow variable time scales would provide additional flexibility for innovating such algorithms. In this paper, we devise a method to extend the application of RB-SIPAs to degraded speech signals with time scale modified relative to the clean reference signal. This is

achieved by: (I) estimating the unknown time-scale change incurred by the degraded signal and (II) incorporating the estimate to modify the RB-SIPA processing steps.

For (I), we estimate the time-scale change between the clean reference signal and the degraded signal, using dynamic time warping (DTW) [12, 13]. DTW finds a correspondence (see “alignment path” defined in Section 2.2.1) between the time frames of the clean and degraded signal by matching the most similar time frames of the two signals. However, in the case of highly degraded speech, finding an accurate alignment path becomes challenging. Traditional audio features, such as the mel frequency cepstral coefficients (MFCCs) [14], perceptual linear prediction (PLP) [15, 16], and linear predictive coding coefficients (LPCCs) [17] degrade with decreasing signal-to-noise ratio (SNR), making it challenging for DTW to find an accurate alignment path [18–22]. In King et. al [23], a probabilistic latent component analysis (PLCA) framework is used to model spectrograms as probability distributions and learn speech components for DTW using expectation-maximization. PLCA features for DTW have shown better alignment accuracy compared to MFCCs, PLP and LPCCs. However, the method relies on expectation-maximization, making it sensitive to initialization. Moreover, the learned components do not have a clear correspondence with perceptually meaningful speech attributes. In this paper, we introduce spectro-temporal modulation (STM) analysis (see Section 2.1) to identify and isolate perceptually interpretable speech components that demonstrate better alignment accuracy compared to MFCCs for various types of noise and SNR levels.

For (II), we propose two methods to incorporate time alignment into existing RB-SIPAs: (a) use the estimated alignment path to align the speech signal spectral sequences calculated by the RB-SIPA, without changing the rest of the RB-SIPA, and (b) in addition to (a), average over the intermediate spectral similarity scores across time indices informed by the alignment path to form the aggregate score output by the RB-SIPA. We demonstrate the generality and efficacy of the proposed method by applying it to four existing RB-SIPAs: Short-Term Objective Intelligibility (STOI) [1], Extended STOI (ESTOI) [2], Spectro-Temporal Glimpsing Index (STGI) [3], and Weighted Spectro-Temporal Modulation Index (WSTMI) [4].

2. Proposed Method

2.1. Spectro-temporal Modulation Analysis

The STM analysis framework provides a computational model for central auditory processing [24, 25]. The model includes two processing stages: (I) calculating log mel spectrogram, and (II) STM decomposition. In the second stage, a Gabor STM filter bank, representing the collection of spectro-temporal receptive

fields, is applied to the log mel spectrogram. Each filter in the filter bank is tuned to a specific spectral and temporal modulation frequency. Together, these filters span a range of STM frequencies that are germane to auditory perception [4, 26].

In this study, we use $S = 11$ spectral modulation frequencies and $R = 4$ temporal modulation frequencies (see Table 2 in [27]) to generate $U = S \times R = 44$ STM frequency channels, i.e., STM-filtered spectrograms that we analyze as features for DTW alignment. The choice of STM frequencies is motivated by prior studies [4, 28] on salience of certain STM channels to SI. Our aim is to determine a subset of the U STM frequency channels that gives the best DTW alignment accuracy for various types of noise. For this purpose, each of the U STM channels is used as a distinct feature representation for finding DTW alignment path. Based on this initial analysis, we combine the best performing STM channels and evaluate their joint impact on DTW alignment accuracy.

2.2. DTW using STM-Decompositions

Let $x[n]$ and $y[n]$ denote a clean and the corresponding time-modified degraded speech signal with log mel spectrograms, $X^{mel}[f, t_x]$, and $Y^{mel}[f, t_y]$ respectively, where $f \in \{1 \dots F\}$ index the acoustic frequency bins, $t_x \in \{1 \dots T_x\}$ and $t_y \in \{1 \dots T_y\}$ index the Short-Time Fourier Transform (STFT) frames of $x[n]$ and $y[n]$, respectively. A log mel spectrogram is decomposed into U STM channels, each denoted by $X^{[s,r]}[f, t_x]$ for $x[n]$ and $Y^{[s,r]}[f, t_y]$ for $y[n]$, where $s \in \{1 \dots S\}$ and $r \in \{1 \dots R\}$ in $[s, r]$ index the spectral and temporal modulation frequencies, respectively.

2.2.1. DTW Alignment Paths

DTW produces an alignment path, $p = \{(p_x^l, p_y^l) \mid l = 1, \dots, L\}$ where (p_x^l, p_y^l) are the STFT frame indices of the two sequences being aligned and L is the length of the path. By definition [12, 13], the path p is continuous, monotonic, and spans the full length of both sequences. To find the alignment paths, we use a computationally optimized version of DTW, known as FastDTW [13] with Euclidean distance metric. Previous studies [20, 23] have shown that using MFCCs as features for DTW yields accurate alignment at high SNR levels, making them suitable for generating reference alignment paths. Therefore, for this study, we leverage datasets where noise-free versions of degraded time-modified speech signals are available, and generate our initial reference alignment paths as follows:

$$p_c^{mfcc} = \text{FastDTW}(X^{mfcc}[m, t_x], \hat{X}^{mfcc}[m, t_y]), \quad (1)$$

where $m \in \{1, \dots, 15\}$ indexes the MFCCs, $X^{mfcc}[m, t_x]$ denotes the MFCCs of clean speech and $\hat{X}^{mfcc}[m, t_y]$ denotes the MFCCs of the noise-free time-modified speech. We call p_c^{mfcc} the *clean path*. Similarly, we denote the *degraded path* as p_d^{mfcc} , obtained using $X^{mfcc}[m, t_x]$ and $Y_\nu^{mfcc}[m, t_y]$, where $Y_\nu^{mfcc}[m, t_y]$ refers to MFCCs of time-modified speech degraded by noise $\nu \in \mathcal{N}$, where $\mathcal{N}$ is a set of different types of noise. In the same way, for each of the U STM channels, we apply FastDTW to STM channel $X^{[s,r]}[f, t_x]$ and $Y_\nu^{[s,r]}[f, t_y]$ to obtain *degraded path* $p_d^{[s,r]}$ (see Equation (1)).

2.2.2. Path Comparison

To evaluate the similarity between two alignment paths, we adopt a threshold-based metric described in [29], which has also been employed in other studies [23, 30]. A frame in a path is

labeled as correct if it lies within τ frames of the corresponding alignment frame in the other path. The percentage agreement (PA) between two paths is determined by the percentage of frames labeled as correct.

2.2.3. Feature Selection

To enhance the DTW alignment performance and improve computational efficiency, we select a subset of the U STM channels based on their average PA scores. Let the PA between p_c^{mfcc} and $p_d^{[s,r]}$ for STM frequency $[s, r]$ be $PA_{[s,r]}^\nu$. Before we compare the PA scores for the STM channels, we perform z-score normalization of the U PA scores for each type of noise $\nu \in \mathcal{N}$:

$$Z_{[s,r]}^\nu = \frac{PA_{[s,r]}^\nu - \mu^\nu}{\sigma^\nu}, \quad (2)$$

where

$$\mu^\nu = \frac{1}{M} \sum_{s=1}^S \sum_{r=1}^R PA_{[s,r]}^\nu, \quad \sigma^\nu = \sqrt{\frac{1}{M} \sum_{s=1}^S \sum_{r=1}^R (PA_{[s,r]}^\nu - \mu^\nu)^2}.$$

Next, we compute the average normalized PA scores $Z_{[s,r]}$ for each STM frequency $[s, r]$ across $|\mathcal{N}|$ types of noise:

$$Z_{[s,r]} = \frac{1}{|\mathcal{N}|} \sum_{\nu \in \mathcal{N}} Z_{[s,r]}^\nu. \quad (3)$$

In the next step, we sort the STM frequencies in descending order by their corresponding STM channels PA scores $Z_{[s,r]}$:

$$\{[s, r]_1, \dots, [s, r]_u, \dots, [s, r]_{U=S \times R}\} = \text{argsort}\{Z_{[s,r]}\}_{[s,r]}. \quad (4)$$

Finally, we select the top K STM channels with nearly identical average normalized PA scores: $\{X^{[s,r]_1}, \dots, X^{[s,r]_K} \mid (Z_{[s,r]_K} - Z_{[s,r]_1}) \leq \varepsilon\}$ where ε is a small positive number. The final STM representation for speech signal $x[n]$ is constructed by stacking the corresponding K STM channels $X^{[s,r]_\nu}$ along the frequency dimension, as follows:

$$X^K[t_x] = \begin{bmatrix} X^{[s,r]_1}[:, t_x] \\ X^{[s,r]_2}[:, t_x] \\ \vdots \\ X^{[s,r]_K}[:, t_x] \end{bmatrix} \in \mathbb{R}^{KF \times 1}. \quad (5)$$

Similarly, we find $Y_\nu^K[t_y]$ for time-modified speech degraded by noise ν . Next, the degraded path $p_d^{[s,r]_K}$ is derived by applying FastDTW [13] to $X^K[t_x]$ and $Y_\nu^K[t_y]$ (see Equation 1).

2.3. Time Alignment

Consider a DTW alignment path $p = \{(p_x^l, p_y^l), l = 1, \dots, L\}$ between clean speech $x[n]$ and time-modified speech $y[n]$, where the (x, y) coordinates of the path represent the correspondences between the time frames of both signals. The alignment is performed on the time-frequency (TF) features, calculated by the RB-SIPA to find SIP. For STOI and ESTOI, the TF-features are the output of their one-third octave band analysis, and for STGI and WSTMI, the TF-features are the result of their STM analysis. Let the TF-features generated by an RB-SIPA for $x[n]$ and $y[n]$ be $X[f_s, t_x]$ and $Y[f_s, t_y]$, respectively, where $f_s \in \{1 \dots F_s\}$ index the frequency representations unique to each RB-SIPA. The corresponding time-aligned TF-features denoted as $X_a[f_s, t]$ and $Y_a[f_s, t]$ are generated using alignment path p as shown in Algorithm 1.

Next, we introduce two modifications to incorporate time alignment into RB-SIPAs: (I) Simple-SIP and (II) Mean-SIP.

Algorithm 1 Time-alignment

Input: $X[f_s, t_x], Y[f_s, t_y]$,
 where $1 \leq f_s \leq F_s, 1 \leq t_x \leq T_x$, and $1 \leq t_y \leq T_y$,
 $p = \{(p_x^l, p_y^l), l = 1, \dots, L\}$
Output: $X_a[f_s, t], Y_a[f_s, t]$
 1: *Initialize:* $X_a \leftarrow \text{zeros}(F, L), Y_a \leftarrow \text{zeros}(F, L)$
 2: **for** $l \leftarrow 1$ **to** L **do**
 3: $X_a[:, l] \leftarrow X[:, p_x^l]$
 4: $Y_a[:, l] \leftarrow Y[:, p_y^l]$
 5: **end for**

2.3.1. Simple-SIP

After performing the time alignment of the TF-features of the RB-SIPA using Algorithm 1, RB-SIPA calculates similarity scores between the time-aligned TF-features $X_a[f_s, t]$ and $Y_a[f_s, t]$. Both TF-features and similarity scores are specific to an RB-SIPA. The time dimension of the aligned TF features is preserved. Simple-SIP retains the RB-SIPA's original way of calculating and combining intermediate similarity scores to produce the SI prediction output.

2.3.2. Mean-SIP

As with Simple-SIP, Mean-SIP calculates the similarity scores between the time-aligned TF-features. Next, Mean-SIP averages the similarity scores across the indices of the alignment path that are repeated due to time scale modification. As a result, repeated frames are represented by a single average score. After this step, Mean-SIP continues as in the original RB-SIPA.

3. Experiments

We divide our experiments into two sections: (I) Experiment-I analyzes STM channels as features for DTW under noise conditions from the UrbanSound8k dataset [31], (II) Experiment-II analyzes the impact of time alignment produced by DTW using STM features identified in Experiment-I, on SIP performance using Simple-SIP and Mean-SIP for Hurricane Challenge-I (HC-I) dataset [5].

3.1. Datasets

3.1.1. Hurricane Challenge-I (HC-I)

The Hurricane Challenge-I [5] was designed to evaluate the performance of various near-end listening enhancement (NELE) algorithms aimed at increasing SI while listening in noisy environments. These pre-degradation SE algorithms were evaluated using listening test scores for SI. The dataset consists of clean speech and corresponding degraded speech produced by two types of additive noise at three different SNRs: (I) competing speech (CS) at -7 dB, -14 dB and -21 dB, and (II) speech-shaped noise (SSN) at 1 dB, -4 dB and -9 dB. For this study, we only test the ten NELE algorithms that modified the time scale of the clean speech signal before adding noise. This provides $2 \times 3 \times 10 = 60$ mean SI scores for the RB-SIPA evaluation.

3.1.2. UrbanSound8k

UrbanSound8k dataset [31] contains urban sounds from ten classes: air conditioner, car horn, dog bark, children playing, drilling, engine idling, gun shot, jackhammer, siren, and street music. Each of these ten types of noise is used to evaluate DTW performance by adding the noise to the clean speech signals from HC-I at 3 different SNRs: -5 dB, -10 dB and -15 dB.

3.2. Experiment-I

Experiment-I analyzes STM channels as features for DTW by comparing *clean paths* with *degraded paths* for ten types of noise from UrbanSound8k [31]. We compare two alignment paths by finding their PA within the threshold $\tau = 2$ which corresponds to 25 ms using the window size of 25 ms with 50% hop-length (see Section 2.2.2).

3.2.1. Evaluation Results-I

Figure 1. shows average PA scores across ten types of noise from UrbanSound8k at three SNR levels: high, medium and low, comparing p_c^{mfcc} with each $p_d^{[s,r]}$. It can be observed that the PA scores corresponding to the STM frequencies: $([s, r]_1, [s, r]_2, [s, r]_3, [s, r]_4) = [(2.88, 4.9), (1.77, 4.9), (2.88, 7.8), (1.77, 7.8)](\text{cyc/kMel}, \text{Hz})$ are the highest, with around 83%, 79% and 65% agreement with p_c^{mfcc} at high, medium and low SNRs, respectively.

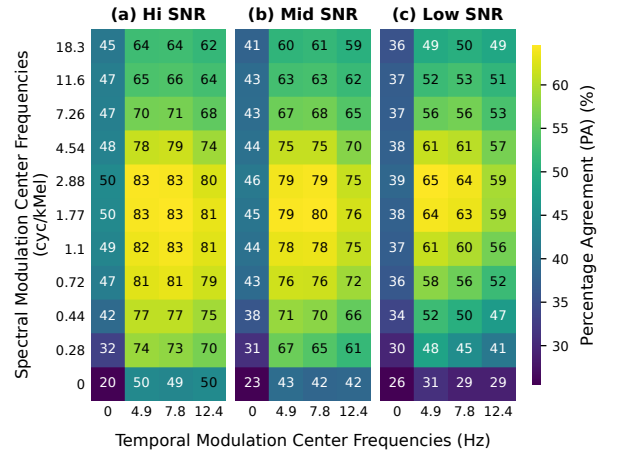


Figure 1: Average PA scores across ten types of noise from UrbanSound8k, calculated between p_c^{mfcc} and $p_d^{[s,r]}$ using threshold $\tau = 2$ (25.6 ms) for high, medium and low SNRs.

Table 1 shows the average PA scores between p_c^{mfcc} and p_d^{mfcc} across ten types of noise from UrbanSound8k [31]. It can be observed that the average PA scores at high, medium, and low SNRs are 78%, 69%, and 52%, respectively, lower than the highest scores in Figure 1. Table 1 also shows the average PA scores using combination of top performing STM channels. It can be observed that *degraded paths* $p_d^{[s,r]K}$ for $K = 1, 2, 3, 4$, show better average agreement with p_c^{mfcc} as compared to *degraded paths* p_d^{mfcc} . PA improves the most by increasing $K = 1$ to $K = 2$. Increasing K beyond 2 yields a minuscule gain in PA score.

Table 1: Average PA scores across ten types of noise from UrbanSound8K, between p_c^{mfcc} and p_d using threshold $\tau = 2$ (25.6 ms) for high, medium and low SNRs.

	p_d^{mfcc}	$p_d^{[s,r]_4}$	$p_d^{[s,r]_3}$	$p_d^{[s,r]_2}$	$p_d^{[s,r]_1}$
Hi	78	86	85	85	83
Mid	69	83	83	82	79
Low	52	68	68	68	65

Table 2: Comparison of listening test scores with Simple-SIP and Mean-SIP incorporating clean path p_c^{mfcc} , degraded path using MFCCs p_d^{mfcc} , and degraded path using STM channels combinations, $p_d^{[s,r]K}$ for $K = 2, 3$ (see Section 3.3.2).

	p_c^{mfcc}				p_d^{mfcc}				$p_d^{[s,r]3}$				$p_d^{[s,r]2}$			
	Simple-SIP															
	STGI	ESTOI	WSTMI	STOI	STGI	ESTOI	WSTMI	STOI	STGI	ESTOI	WSTMI	STOI	STGI	ESTOI	WSTMI	STOI
Pear	0.77	0.88	0.89	0.51	0.64	0.67	0.71	0.42	0.82	0.90	0.92	0.54	0.81	0.89	0.92	0.53
Spear	0.76	0.88	0.88	0.52	0.64	0.67	0.69	0.42	0.81	0.89	0.92	0.54	0.81	0.89	0.91	0.52
CCC	0.73	0.86	0.87	0.41	0.57	0.61	0.66	0.30	0.78	0.88	0.91	0.44	0.78	0.87	0.90	0.42
Ken	0.56	0.70	0.69	0.35	0.44	0.48	0.50	0.27	0.63	0.71	0.74	0.37	0.63	0.71	0.73	0.36
RMSE	0.16	0.11	0.11	0.21	0.19	0.18	0.17	0.22	0.14	0.11	0.09	0.20	0.14	0.11	0.10	0.21
	Mean-SIP															
Pear	0.79	0.89	0.90	0.55	0.70	0.69	0.74	0.52	0.85	0.91	0.94	0.60	0.85	0.92	0.93	0.59
Spear	0.78	0.89	0.90	0.55	0.69	0.67	0.73	0.51	0.84	0.91	0.93	0.61	0.84	0.92	0.93	0.60
CCC	0.75	0.88	0.89	0.45	0.65	0.63	0.69	0.41	0.82	0.90	0.92	0.51	0.82	0.90	0.92	0.50
Ken	0.59	0.72	0.71	0.38	0.49	0.49	0.52	0.33	0.66	0.75	0.77	0.44	0.65	0.75	0.77	0.43
RMSE	0.15	0.11	0.10	0.20	0.17	0.18	0.16	0.21	0.13	0.09	0.09	0.20	0.13	0.09	0.09	0.20

3.3. Experiment-II

Experiment-II involves using STM channels identified in Section 3.2 to find *degraded paths* and incorporating them into RB-SIPAs to study the impact of alignment on SIP performance using Simple-SIP and Mean-SIP.

3.3.1. Evaluation Metrics

Simple-SIP and Mean-SIP are used to find SIP scores for four existing RB-SIPAs (see Section 1) which are compared with listening test SI scores in HC-I, using five metrics: (I) Pearson correlation coefficient (Pear), (II) Spearman rank correlation coefficient (Spear), (III) Concordance correlation coefficient (CCC), (IV) Root mean square error (RMSE) and (V) Kendall's correlation coefficient (Ken).

3.3.2. Evaluation Results-II

Table 2. displays the SIP performance on HC-I for four RB-SIPAs using *clean paths* and *degraded paths*. SIP metric scores in the column p_c^{mfcc} after incorporating *clean path* p_c^{mfcc} (see Section 2.3), providing us with baseline SIP metric scores. Similarly, the columns p_d^{mfcc} , $p_d^{[s,r]3}$ and $p_d^{[s,r]2}$ present the SIP metric scores after incorporating time alignment using *degraded paths* p_d^{mfcc} , $p_d^{[s,r]3}$ and $p_d^{[s,r]2}$, respectively. The STM channels used to calculate the degraded paths are described in Section 3.2. For all four RB-SIPAs, the SIP performance using p_d^{mfcc} decreases relative to the baseline SIP metric scores. On the other hand, incorporating $p_d^{[s,r]3}$ and $p_d^{[s,r]2}$ show better SIP performance than the baseline SIP metric scores in column p_c^{mfcc} . SIP performance does not improve further if more than three STM channels are selected for time alignment. Compared with MFCC-based alignment, the SIP performance of STM-based alignment suggests that STM features are more robust to noise degradation, leading to a more accurate alignment path and improved SIP performance.

4. Discussion

The experimental results show that the range of STM channels important for the time alignment of speech signals overlaps with those essential for SI, i.e., spectral modulation frequencies below ~ 4 cyc/kMel and temporal modulation frequencies below ~ 8 Hz [4, 28]. This range of spectral modulation frequencies is occupied by energy associated with the resonant frequencies of the vocal tract, i.e., formants, which are crucial for differentiating vowels and some consonant sounds. Also,

the temporal modulation frequencies below 8 Hz lies within the temporal modulations range corresponding to the conversational rate of syllables, at around 2 to 6 Hz [32, 33]. The regions within these modulation frequency ranges over which they disagree comprise spectral frequencies below ~ 0.2 cyc/kMel and temporal frequencies below ~ 2 Hz. This is evident given that the regions of low spectro-temporal modulation frequencies lack the spectro-temporal details to help DTW find an accurate alignment path. Therefore, the STM channels salient for SI but lack sufficient spectro-temporal details are ineffective for precise speech alignment. Thus, for time alignment, our scheme deploys SI-salient STM channels that also preserve critical spectro-temporal details in the presence of noise.

5. Conclusion

In this paper, we propose a novel approach leveraging STM features for DTW to achieve accurate time alignment between clean and degraded speech signals. We identify noise-robust STM channels across various types of noise and present a method to select multiple STM channels to improve time alignment. Our results demonstrate that DTW using STM features shows better alignment accuracy compared to DTW using MFCCs features. Furthermore, we demonstrate the integration of the time alignment into RB-SIPAs and use the augmented RB-SIPAs to generate SIP scores. We further compare the output of RB-SIPA with listening test scores and show superior correlation performance using STM features compared to MFCCs. Our findings reveal that the STM channels critical for the time alignment of speech overlap with those essential for SI.

6. References

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